

# Apple Inc. (AAPL) Historical Stock Market Analysis

## Insight Summary Report

AnalystLab Africa Internship – Week 6

### Executive Summary

This report presents an analysis of five years of Apple Inc. (AAPL) historical stock market data using advanced Python techniques. The project focused on data transformation, time-series analysis, feature engineering, and visualization to identify trends in Apple's stock performance. Python libraries including Pandas, NumPy, and Matplotlib were used to clean, transform, analyze, and visualize the dataset.

The analysis revealed strong long-term growth in Apple's stock price, accompanied by moderate market volatility and consistently high trading activity. Feature engineering techniques such as moving averages, daily returns, monthly returns, and volatility measures provided additional insights into market behavior and investment trends.

---

## Dataset Overview

The dataset consists of five years of Apple Inc. historical stock market data obtained through Yahoo Finance. Each record represents one trading day and includes the following variables:

- Date
- Open Price
- High Price
- Low Price
- Close Price
- Adjusted Close Price
- Trading Volume

These variables provide the foundation for analyzing historical stock performance, trading activity, and market trends.

---

## Data Transformation

Several preprocessing and transformation techniques were applied to prepare the dataset for analysis.

The Date column was converted into a datetime format to enable time-series operations. Additional features were engineered from the original variables, including:

- Daily Price Change

- Daily Percentage Change
- High-Low Trading Range
- Year
- Month
- Day of the Week

These derived features enhanced the analytical capabilities of the dataset and supported more detailed trend analysis.

---

## Time-Series Analysis

Time-series analysis was performed to evaluate Apple's stock performance over the five-year period.

The analysis included:

- Closing Stock Price Trend
- Trading Volume Trend
- 7-Day Moving Average
- 30-Day Moving Average
- Monthly Returns Analysis

The moving averages smoothed short-term fluctuations and highlighted the long-term upward trend of Apple's stock price. Monthly return analysis further illustrated periods of positive and negative market performance throughout the observation period.

---

## Feature Engineering

Additional analytical features were created to better understand market behaviour.

These included:

- Daily Return
- Monthly Return
- 7-Day Moving Average
- 30-Day Moving Average
- Rolling 30-Day Volatility

These engineered features provided valuable insights into price momentum, market stability, and overall stock performance.

---

## Key Findings

The analysis produced several notable findings:

- Average Closing Stock Price: **\$196.61**

- Average Daily Trading Volume: **65,084,442 shares**
- Average Daily Return: **0.08%**
- Average 30-Day Volatility: **1.64%**
- Overall Stock Growth (Five Years): **117.49%**

The results indicate that Apple experienced substantial long-term growth despite periods of market volatility. Trading volume remained consistently high throughout the analysis period, reflecting strong investor interest and liquidity.

---

## Recommendations

Based on the analysis, the following recommendations are proposed:

- Continue monitoring long-term price trends using moving averages to identify market direction.
  - Incorporate volatility measures when assessing investment risk.
  - Combine technical indicators with external market and economic factors for more informed investment decisions.
  - Extend future analyses by comparing Apple with other major technology companies to evaluate relative market performance.
- 

## Conclusion

This project successfully demonstrated the application of advanced Python techniques for financial data analysis. Through data transformation, feature engineering, and time-series analysis, meaningful insights were extracted from five years of Apple Inc. historical stock data.

The findings showed that Apple's stock achieved approximately **117.49% overall growth** during the study period while maintaining strong trading activity and moderate volatility. The project highlights how Python can be used to transform raw financial data into actionable insights that support business intelligence and investment decision-making.